

每日一題36

單元7 三角比-直線的斜角

2025.10.06

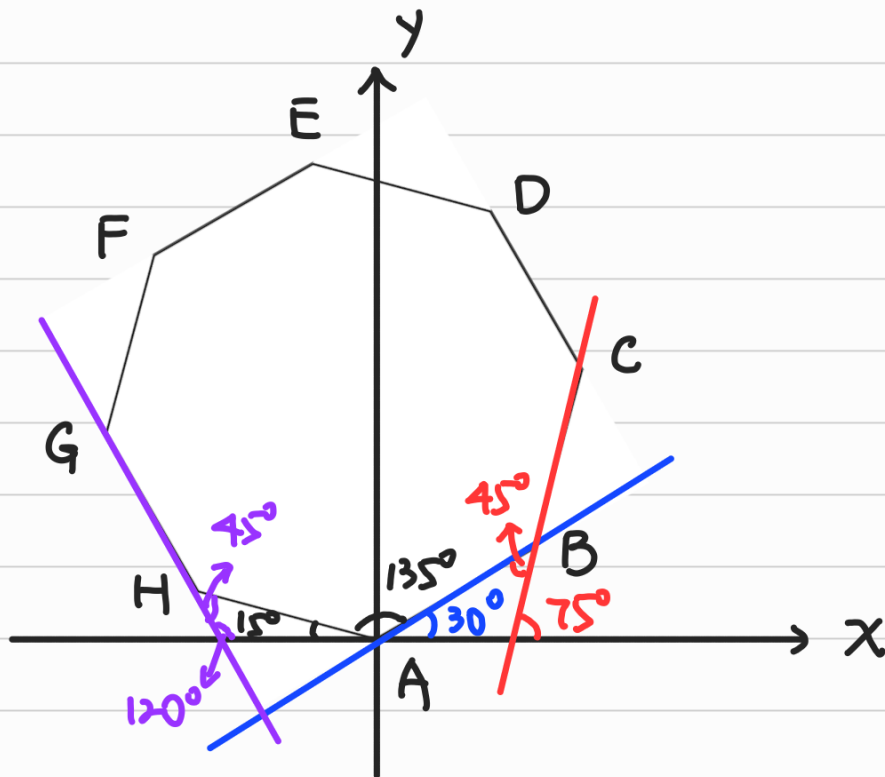
114南一第=次模考#5

坐標平面上正八邊形的其中一邊落在斜率為 $\frac{1}{\sqrt{3}}$ 的直線上，若此正八邊形每個邊所在直線的斜率最大值為 α 、最小值為 β ，試問 $(1-\alpha)(1+\beta)$ 之值為何？（ $\tan 75^\circ = 2 + \sqrt{3}$ ）

- (1) -2 (2) $-\sqrt{6}-\sqrt{2}$ (3) $\sqrt{3}$
 (4) $\sqrt{6}-\sqrt{2}$ (5) 2

<Sol>

$$\tan \theta = m = \frac{1}{\sqrt{3}} \Rightarrow \theta = 30^\circ$$



$$m_{AB} = m_{EF} = \tan 30^\circ = \frac{1}{\sqrt{3}}$$

$$m_{AH} = m_{DE} = \tan 15^\circ = \frac{1}{\tan 75^\circ} = \frac{1}{2 + \sqrt{3}} = 2 - \sqrt{3}$$

$$m_{BC} = m_{FG} = \tan 75^\circ = 2 + \sqrt{3}$$

$$m_{CD} = m_{GH} = \tan 120^\circ = -\tan 60^\circ = -\sqrt{3}$$

$$\therefore \text{Max} = \alpha = 2 + \sqrt{3} \cdot \text{min} = \beta = -\sqrt{3}$$

$$\Rightarrow (1 - \alpha)(1 + \beta) = [1 - (2 + \sqrt{3})][1 + (-\sqrt{3})]$$

$$= (-1 - \sqrt{3})(1 - \sqrt{3}) = -(1 + \sqrt{3})(1 - \sqrt{3})$$

$$= -[1^2 - (\sqrt{3})^2] = -(1 - 3) = -(-2) = 2 \quad \#$$